PRODUCT USER MANUAL

Atlantic - European North West Shelf - Ocean Biogeochemistry Analysis and Forecast Product

NWSHELF_ANALYSISFORECAST_BGC_004_002

Issue: 3.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2020/08	All	Merged PUM_004_001 and PUM_004_002. Updated for Dec 2020 release.	M. Tonani and I. Ascione	
1.1	2021/01	All	Updates for May 2021 release	M. Tonani and I. Ascione	
1.2	2021/08	II.3 and II.4	Details on vertical grid and vertical interpolation.	M. Tonani and I. Ascione	
1.3	2021/01	II.2 and II.4	Upgrade to GLO-HR and new MDT	A.Saulter, I. Ascione	C. Derval
2.0	2023/06/16	All	**[EIS November 2023]** First release of the NWS biogeochemical analysis and forecast product, as an extraction of the IBI operational system.	A. Amo, E. Gutknecht, L. Castrillo	M.G. Sotillo, S. Cailleau
2.1	2024/05/31	All	**[EIS November 2024]** New variable distributed: light attenuation.	L. Castrillo, R. Aznar	S. Cailleau, M. G. Sotillo
3.0	2025/05	All	**[EIS November 2025]** Re-introduction of the NWS biogeochemical analysis and forecasting product by the UK Met Office	J.Tinker, S.Kay, T.Brüning	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NWS	North-West Shelf
TAC	Thematic Assembly Center
GLO	Global
IOC	Intergovernmental Oceanographic Commision of UNESCO
NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
OPENDAP	Open-Source Project for a Network Data Access Protocol. Protocol to download subset of data from a n-dimensional gridded dataset (ie: 4 dimensions: lon-lat, depth, time)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	A hindcast done without data assimilation.
ANALYSIS	A hindcast adjusted with data assimilation

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the North West-European Shelf analysis and forecast product: **NWSHELF_ANALYSISFORECAST_BGC_004_002.**

This product is generated by a coupled physics-biogeochemistry forecasting system. The corresponding physical product is NWSHELF_ANALYSISFORECAST_PHY_LR_004_001.

Every day:
- the time series (analysis for 2 days before present and 7 days forecast) is updated at the end of the operational production process
- old forecasts are deleted from the MDS every day

The products are published on and available through the MDS after automatic and human quality controls. Downloaded files are in NetCDF format and follow CF-1.7 convention.

The analysis and forecasting system is described in the Quality Information Document (QUID): https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-002.pdf

More detailed information can be obtained from the Copernicus Marine Service Team.

b) History of changes

Date	Description of changes and impacted product
December 2020	Re-delivery of the historical hindcast time series. Addition of total Chlorophyll data assimilation.
May 2021	Product / dataset name change.
February 2022	Boundary conditions have been upgraded
November 2023	First release of the NWS biogeochemical analysis and forecast product, as an extraction of the IBI operational system.
November 2024	Distribution of diffuse attenuation coeff of light in sea water, averaged over the 400-500 nm range (light attenuation).
November 2025	Re-introduction of the NWS biogeochemical analysis and forecasting product by the UK Met Office including a 7-day forecast

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

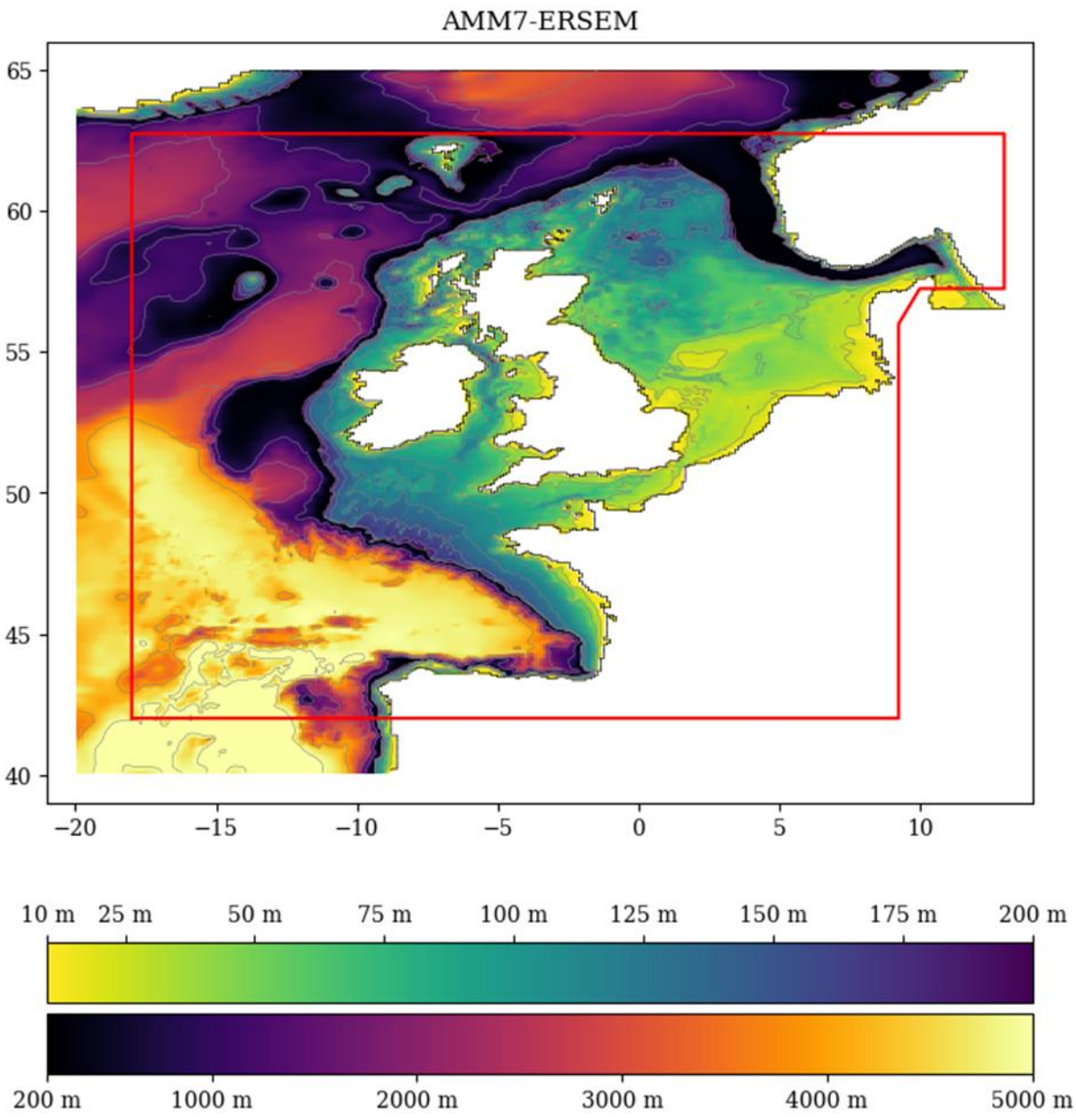
a) General Information

Product Lines	NWSHELF_ANALYSISFORECAST_BGC_004_002	
Geographical coverage	20°W -> 13°E; 40°N -> 65°N. The outermost model grid points at the Atlantic and Baltic boundaries are masked. Baltic boundary: all the grid points in the Baltic and the Kattegat South of 57.25°N are masked. Atlantic boundaries: grid points north of 64°18'48" N; west of -18°19'12" E and south of 40°43'58" N are masked (outermost 10-grid points).	
Variables	• Volume Beam Attenuation Coefficient of Radiative Flux • Concentration of Chlorophyll • Mole Concentration of Dissolved Oxygen • Mole Concentration of Nitrate • Mole Concentration of Phosphate • Mole Concentration of Phytoplankton Expressed as Carbon • Net Primary Production • Surface partial pressure of carbon dioxide • pH reported on total scale	
Product Type	Near Real Time - Analysis	Near Real Time - Forecast
Available time series	Two years before present	7 days forecast
Temporal resolution	Daily mean	Daily mean
Update Frequency	Daily	Daily
Target delivery time	daily at 12 pm (noon) UTC	
Horizontal resolution	1/9° longitude x 1/15° latitude	
Number of vertical levels	24	
Format	NetCDF4 – CF1.7	
Delivery mechanism	Copernicus Marine Toolbox	

b) Production System Description

The operational North-West-European Shelf analysis and forecast products are generated by a coupled physical-biogeochemical forecasting system at 1/15° (longitude) and 1/9° (latitude). It is providing 7 days of 2D/3D (PHY_LR_004_001) or (BGC_004_002) data updated daily. This product includes daily mean files. Rolling archive of 2 years is available.

NORTH WEST EUROPEAN SHELF (20°W - 13°E; 40°N - 65°N) :
1/15° longitude x 1/9° latitude, regular grid, 297x 375

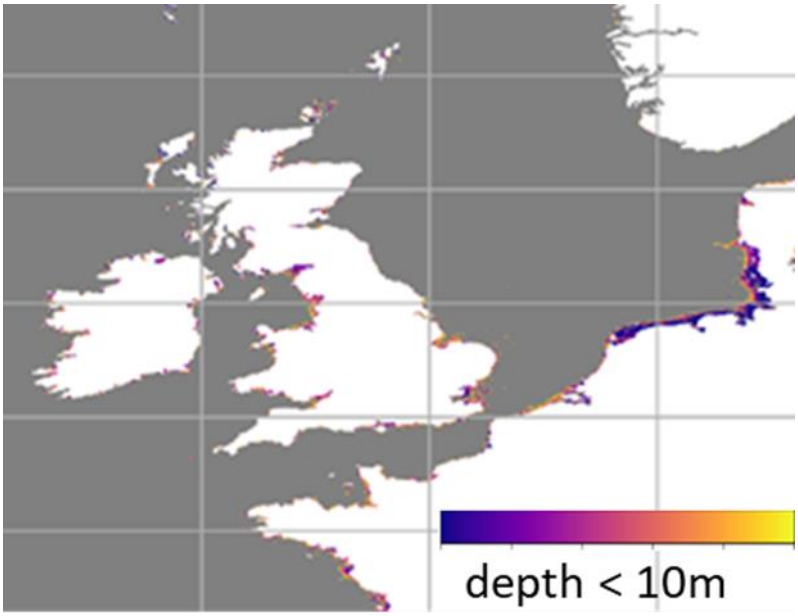


NWS bathymetry (m) of the AMM7 model domain, with the red line showing the product domain.

The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises open waters of the shelf seas, i.e. (using UK terminology) the North Sea, Irish Sea, English Channel, Celtic Sea and Bay of Biscay.

The model is a nested regional ocean model configuration, coupled with the physical model. PHY and BIO products are delivered on an identical horizontal grid.

A **minimum depth of 10m** is imposed in the ocean model for considering the sea level tidal excursion. Products below the surface in areas where the bathymetry is less than 10m could be affected by this modelling choice.



1) What is done:

We take every bathymetric grid point < 10 m in the domain and reassign it with a depth of 10 m, coloured point in the figure above. These grid points are interpolated back to their original depth before delivering the file to the CMEMS catalogue. Areas such as the Wadden Sea are particularly affected (see below reference for a case study in the Wadden sea).

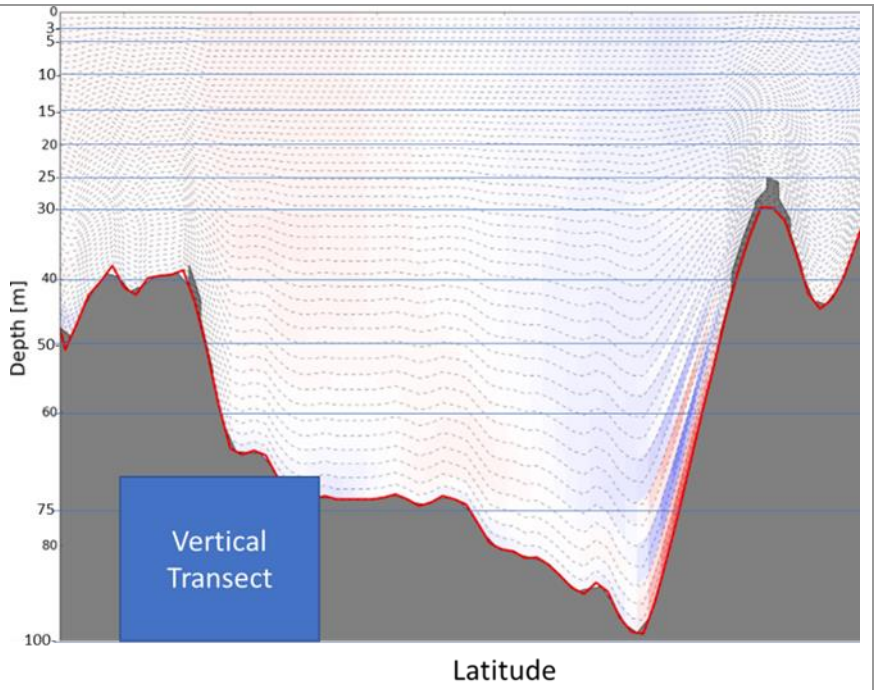
2) Why it is done:

This is because version 3.6 of NEMO does not have wetting and drying capability. If the bathymetry were not modified, it would mean that with the large tides that exist in this domain the model would not be able to handle the case when the tide goes out.
The minimum depth prevents such a situation from occurring and keeps the model as a whole stable.

3) What is the likely effect.

The prorogation of tides and surges are inherently related to the water depth. The speed is proportional to $\sqrt{gH}$ where H is the water depth and thus as the model water depth is deeper than reality the prorogation will be incorrect.
In cases where the seabed is exposed then this process is completely absent from the model by design (there will be water in the model where in reality there is none). It is thus not recommended to the user to directly use the model fields in regions where there are extensive areas of bathymetry less than 10 m in depth. If a model is to be nested within this product, then it is recommended that the boundaries of the nested domain are sufficiently far from such areas to reduce their impact.

4) Wetting and Drying is implemented in NEMO 4. This specific problem is addressed with the wetting and drying module in NEMO 4 which will be part of future development of this operational Copernicus product. The interested reader is directed to the recent paper on its development: https://doi.org/10.1016/j.ocemod.2020.101708 Enda O'Dea, Michael J. Bell, Andrew Coward, Jason Holt, Implementation and assessment of a flux limiter based wetting and drying scheme in NEMO, Ocean Modelling, Volume 155,2020.	
Horizontal grid	~7km (1/9°longitude x 1/15°latitude)
Vertical grid	The products are interpolated for users' convenience, into 24 standard IOC geopotential levels: [surface, 3, 10, 15, 20, 30, 50, 75, 100, 125, 150, 200, 250,300, 400, 500, 600, 750, 1000, 1500, 2000, 3000, 4000, 5000 m]. The surface level is not interpolated, it is the first model level. It has a thickness of 1m where the bathymetry is deeper than 50m. Its thickness is less than 1m where the bathymetry is shallower than 50m. The model native vertical grid has Hybrid S-σ-z-coordinates. The terrain-following coordinates system uses a stretching function to maintain near uniform vertical resolution at the surface. The number of S-σ-z levels is 51 almost everywhere in model domain. It means that the thickness of the model vertical grid cells is smaller in shallow area compare to the deep part of the basin. The thickness of the model native vertical grid varies from 0.3m (in areas with a bathymetry < 50m) to 99m (where the bathymetry is > 4000m).

	 *Vertical transect: it shows the differences between the model native vertical grid (dotted line) and the CMEMS products interpolated at standard geopotential depths (blue lines). The model bathymetry (red lines) doesn't match exactly the bathymetry (shaded grey area) due to numerical constrains.* The bottom temperature (bottomT) is output directly from the native model, it is not interpolated. All other CMEMS NWS-MFC products are vertically interpolated from the 51 native terrain following coordinates to fixed depths.
System Name	AMM7
Model version	ERSEM22.11 (coupled to physical model NEMO 3.6 via FABM)
Coupling	One-way on-line coupling: physics forcing the biogeochemistry. Coupling frequency: at every model time step.
Bathymetry	GEBCO 1 arc-minute data together with a variety of other local data sources, supplied by the North West-European Shelf Operational Oceanographic System (NOOS) partners.
Initial conditions	The model run started on 01/01/2017 from the previous version of NEMO_ERSEM forecasting system.
Atmospheric forcings	Met Office Global Unified Model (MetUM)
Tidal constituents	Tidal forcing is included both on the open boundary conditions via a Flather radiation boundary conditions (Flather, 1976) and through the inclusion of equilibrium tide from a tidal model of the North-East Atlantic (Flather, 1981). 15 tidal constituents: M2, S2, N2, K2, K1, O1, P1, Q1, M4, MS4, L2, T2, S1, 2N2, MU2, NU2.
Rivers	Daily timeseries of river discharge, nutrient loads (nitrate, phosphate, silicate, ammonia), alkalinty (total alkalinity, bioalkalinity, dissolved organic carbon), and oxygen from 1991-2017. Timeseries data are produced from an updated version of the river dataset used in Lenhart et

	al. (2010) combined with climatology of daily discharge data from the Global River Discharge Data Base (Vörösmarty et al., 2000) and from data prepared by the Centre for Ecology and Hydrology as used by Young and Holt (2007). Interannual changes in the series are small and so the 2017 time series is repeated for subsequent years.
Lateral boundary condition	• Atlantic and Baltic (Nitrate, Phosphate and Oxygen): WOA 2023 monthly climatology • Atlantic and Baltic (Alkalinity and dissolved inorganic carbon): GLODAP v2 climatology
Surface boundary condition	• Nitrogen deposition at surface from EMEP project (time-varying).
Data Assimilation scheme	3D Var FGAT. NEMOVAR.
Assimilated observations	• satellite Chlorophyll from CMEMS OC-TAC.

c) Dataset details

NWSHELF_ANALYSISFORECAST_BGC_004_002	
Daily mean (**P1D-m**) fields of: • Volume Beam Attenuation Coefficient of Radiative Flux • Concentration of Chlorophyll • Mole Concentration of Dissolved Oxygen • Mole Concentration of Nitrate • Mole Concentration of Phosphate • Mole Concentration of Phytoplankton Expressed as Carbon • Net Primary Production • Surface partial pressure of carbon dioxide • pH reported on total scale • Total Alkalinity • Mole Concentration of Dissolved Inorganic Carbon • Mole Concentration of Ammonium • Mole Concentration of Silicate • Euphotic Zone Depth	
Datasets	Variable's name in the NetCDF file [Unit] Long_name CF Standard_name
cmems_mod_nws_bgc-**kd**_anfc_7km-3D_P1D-m	**attn** [m-1] Volume Beam Attenuation Coefficient of Radiative Flux in Sea Water volume_beam_attenuation_coefficient_of_radiative_flux_in_sea_water

cmems_mod_nws_bgc-**chl**_anfc_7km-3D_P1D-m	**chl** [mg m-3] Mass concentration of Chlorophyll in Sea Water mass_concentration_of_chlorophyll_a_in_sea_water
cmems_mod_nws_bgc-**o2**_anfc_7km-3D_P1D-m	**o2** [mmol m-3] Mole Concentration of Dissolved Oxygen in Sea Water mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
cmems_mod_nws_bgc-**no3**_anfc_7km-3D_P1D-m	**no3** [mmol m-3] Mole Concentration of Nitrate in Sea Water mole_concentration_of_nitrate_in_sea_water
cmems_mod_nws_bgc-**po4**_anfc_7km-3D_P1D-m	**po4** [mmol m-3] Mole Concentration of Phosphate in Sea Water mole_concentration_of_phosphate_in_sea_water
cmems_mod_nws_bgc-**phyc**_anfc_7km-3D_P1D-m	**phyc** [mmol m-3] Mole Concentration of Phytoplankton Expressed as Carbon in Sea Water mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water
cmems_mod_nws_bgc-**pp**_anfc_7km-3D_P1D-m	**nppv** [mg m-3 day-1] Net Primary Production net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
cmems_mod_nws_bgc-**spco2**_anfc_7km-2D_P1D-m	**spco2** [Pa] Surface partial pressure of carbon dioxide in sea water surface_partial_pressure_of_carbon_dioxide_in_sea_water
cmems_mod_nws_bgc-**ph**_anfc_7km-3D_P1D-m	**ph**[1] Sea Water pH reported on total scale Sea_water_ph_reported_on_total_scale
cmems_mod_nws_bgc-**talk**_anfc_7km-3D_P1D-m	**talk** [mmol m-3] Sea Water alkalinity expressed as mole equivalent sea_water_alkalinity_expressed_as_mole_equivalent
cmems_mod_nws_bgc-**dissic**_anfc_7km-3D_P1D-m	**dissic** [mmol m-3] Mole Concentration of dissolved inorganic carbon in sea water mole_concentration_of_dissolved_inorganic_carbon_in_sea_water
cmems_mod_nws_bgc-**nh4**_anfc_7km-3D_P1D-m	**nh4** [mmol m-3] Mole concentration of ammonium in sea water mole_concentration_of_ammonium_in_sea_water
cmems_mod_nws_bgc-**si**_anfc_7km-3D_P1D-m	**si** [mmol m-3] Mole concentration of silicate in sea water mole_concentration_of_silicate_in_sea_water
cmems_mod_nws_bgc-**zeu**_anfc_7km-2D_P1D-m	**zeu** [m] Euphotic zone depth euphotic_zone_depth
cmems_mod_nws_bgc_anfc_7km-3D_**static**	**mask** Land-sea mask sea_binary_mask
	depth [m] depth depth
	deptho [m] Bathymetry sea_floor_depth_below_geoid
	mdt[m] Mean Dynamic Topography

	sea_surface_height_above_geoid
	deptho_lev_interp deepest wet grid point of the interpolated 3D fields model_level_number_at_sea_floor

The model **bathymetry, Land-Sea mask** and the Mean Dynamic Topography **(MDT)** for this product is the same as for the product NWSHELF_ANALYSISFORECAST_PHY_LR_004_001 as both products are obtained from a coupled physical-biogeochemical model system.

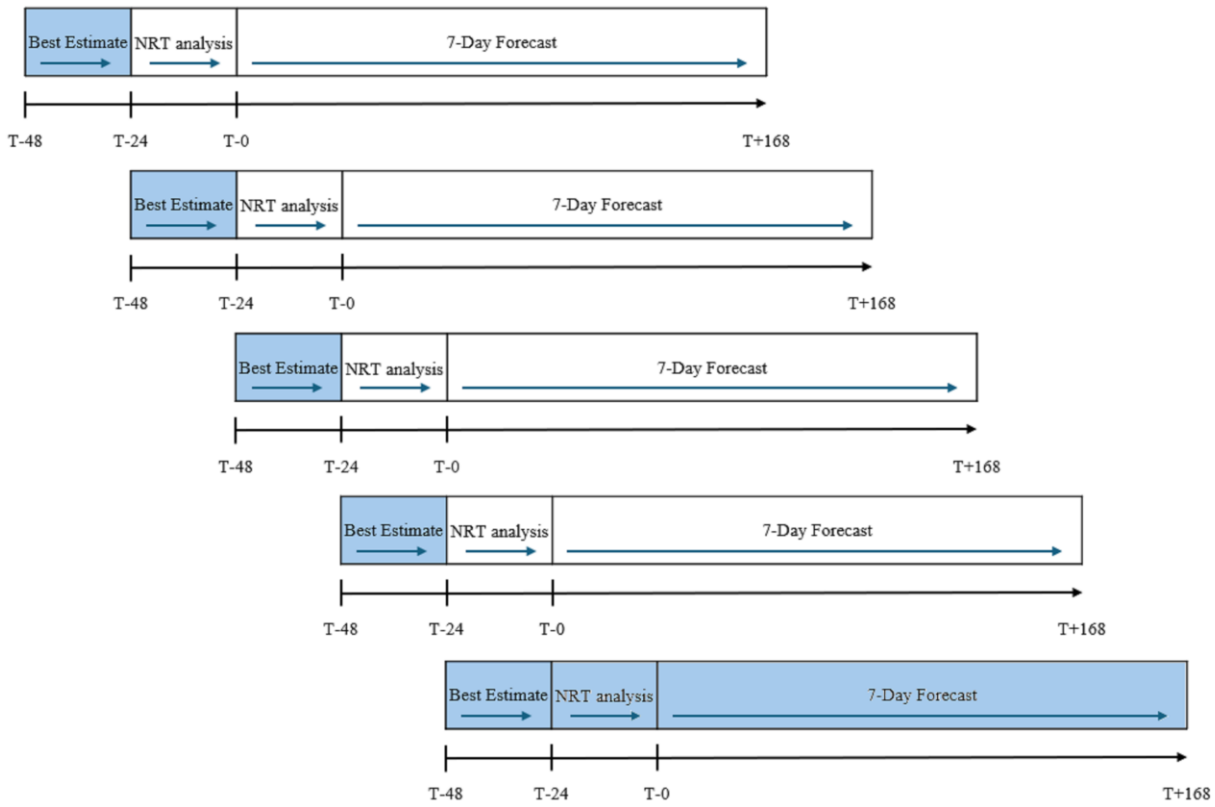
d) Processing information

Product disseminating schedule :

Type of product (forecast/ hindcast)	**Forecast run (base time)**	**Product temporal coverage (hours)**	**Product disseminating scheduled time**	**Dissemination day**
Forecast	00 UTC	0 to 168	until 12 UTC	Every day
NRT - Analysis	00 UTC	-24 to 0	until 12 UTC	Every day
Best estimate - Analysis	00 UTC	-48 to -24	until 12 UTC	Every day

e) Catalogue updating

The time coverage of the catalogue data spans from two years before present to present plus 7 days of forecast.



Each day the [T-48h, T-24h] and [T-24h, T+00h] analyses are provided along with 7 days forecast for this product (with the first day of the forecast being for the day of production).
The delivery usually happens between 6:40 AM and 10:00 AM, a couple of hours the target delivery time. The previous day 'NRT analysis' and '7-day of forecast' are deleted at the beginning of this process. The catalogue data from 2-year before present to present minus 24 hours is taken from T-48h to T-24h of each run (Best Estimate - Analysis).

f) Time averaging

Dataset frequency	Time averaging	Forecast lead time
daily mean values	Values are calculated as means of 25 hours, starting at midnight (UTC) and finishing on the following midnight to remove both diurnal and tidal cycles.	7-day (168-hr)

g) Product use and impact

Specialised marine biogeochemistry analysis and forecast product for the Northwest Shelf region. Uses include marine navigation, emergency response (search and rescue/pollution), coastal flood mitigation, health and ecosystems monitoring, energy and net zero. Applications are developed for government policy, industrial decision making and academic research. The NWS products provide high resolution data and forecasts for ocean state in the tidally dominated waters of the Northwest Shelf and are designed to provide state of the art information specific to this region.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1) FILE SIZE

NWSHELF_ANALYSISFORECAST_BGC_004_002		
DATASET NAME	**NAME OF FILE**	**DIMENSION [MB]**
cmems_mod_nws_bgc-kd_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_ATTN_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.7
cmems_mod_nws_bgc-chl_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_CPWC_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	3.2
cmems_mod_nws_bgc-o2_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_DOXY_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.6
cmems_mod_nws_bgc-no3_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_NITR_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.8
cmems_mod_nws_bgc-po4_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_PHOS_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.9
cmems_mod_nws_bgc-phyc_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_PHYT_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	3.2
cmems_mod_nws_bgc-pp_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_PPRD_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	3.2

cmems_mod_nws_bgc-spco2_anfc_7km-2D_P1D-m	metoffice_foam1_amm7_NWS_PCO2_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	0.3
cmems_mod_nws_bgc-ph_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_PHPH_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.1
cmems_mod_nws_bgc-talk_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_TALK_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.8
cmems_mod_nws_bgc-dissic_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_DISSIC_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.8
cmems_mod_nws_bgc-nh4_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_NH4_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.8
cmems_mod_nws_bgc-si_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_SI_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.8
cmems_mod_nws_bgc-zeu_anfc_7km-2D_P1D-m	metoffice_foam1_amm7_NWS_ZEU_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	0.3
cmems_mod_nws_bgc_anfc_7km-3D_static	NWS-MFC_004_002_static.nc	12

4.2) STRUCTURE OF FILES

An examples of the header of output NetCDF files is inserted in **annex**.

5) REFERENCES

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-002.pdf

Flather R.A., (1976). A tidal model of the north west European continental shelf. Memoires de la Societe Royale de Sciences de Liege,6,141-164.

Flather R.A., (1981). Practical surge prediction using numerical models. In: Peregrine DH (ed) Floods due to high winds and tides. Academic Press, London New York.

Lenhart H.-J., Mills D.K., Baretta-Bekker H., van Leeuwen S.M., van der Molen J., Baretta J.W., Blaas M., Desmit X., Kühn W., Los H.J., Ménesguen A., Neves R., Proctor R., Ruardij P., Skogen M.D., Vanhoutte-Brunier A., Villars M.T., Wakelin S., (2010). Predicting the consequences of nutrient reduction on the eutrophication status of the North Sea, Journal of Marine Systems, Vol 81,148-170.

Vörösmarty J., Green P., Salisbury J., Lammers R. B., (2000). Global Water Resources: Vulnerability from Climate Change and Population Growth, Science, 289, 284–288, https://doi.org/10.1126/science.289.5477.284

Young E.F, Holt J.T., (2007). Prediction and analysis of long-term variability of temperature and salinity in the Irish Sea. Journal of Geophysical Research, 112 (C1). Art. No. C01008. https://doi.org/10.1029/2005JC003386

6) ANNEX

```
netcdf metoffice_foam1_amm7_NWS_CPWC_b20250526_dm20250524 {
dimensions:
        depth = 24 ;
        time = UNLIMITED ; // (1 currently)
        latitude = 375 ;
        longitude = 297 ;
variables:
        double time(time) ;
                time:axis = "T" ;
                time:standard_name = "time" ;
                time:data_time = 86400.f ;
                time:long_name = "Validity time" ;
                time:calendar = "Gregorian" ;
                time:units = "seconds since 1970-01-01T00:00:00Z" ;
        float latitude(latitude) ;
                latitude:axis = "Y" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:nav_model = "grid_T" ;
                latitude:long_name = "latitude" ;
                latitude:valid_min = -90.f ;
                latitude:valid_max = 90.f ;
        float longitude(longitude) ;
                longitude:axis = "X" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:nav_model = "grid_T" ;
                longitude:long_name = "longitude" ;
                longitude:valid_min = -180.f ;
                longitude:valid_max = 180.f ;
        float chl(time, depth, latitude, longitude) ;
                chl:_FillValue = -32768.f ;
                chl:missing_value = -32768.f ;
                chl:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
                chl:long_name = "Concentration of Chlorophyll in Sea Water" ;
                chl:units = "mg m-3" ;
                chl:valid_min = 0.f ;
                chl:valid_max = 795.f ;
        float depth(depth) ;
                depth:axis = "Z" ;
                depth:standard_name = "depth" ;
                depth:units = "m" ;
                depth:positive = "down" ;
                depth:long_name = "depth" ;

// global attributes:
                :Conventions = "CF-1.7" ;
                :title = "daily-mean chlorophyll concentration (3D)" ;
                :references = "http://marine.copernicus.eu/" ;
                :institution = "UK Met Office" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
```

:netcdf-version-id = "netCDF-4" ;
:creation_date = "2025-05-26T07:06:42Z" ;
:source = "PS-OS 47, AMM-FOAM 7 km (tidal) NEMO v3.6_FABM-ERSEM v19.04_NEMOVAR v6" ;
:credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
:licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-licence/" ;
:product = "NWSHELF_ANALYSISFORECAST_BGC_004_002" ;
:history = "" ;
:forcing_data_source = "UKMO Unified Model (MetUM) Atmospheric model Coupled-NWP (applied as direct fluxes); UKMO global ORCA12 ocean model; UKMO GLOWave; CMEMS-BALTIC; climatological rivers" ;
}" ;